
Helix
An electronic digital toolkit


Framework

Function: Alphabet Remover

- Return False when Alphabet is entered

```
bool NumChecker(string inputdata) {
    for (int i = 0; i < inputdata.length(); i++)
        if (isdigit(inputdata[i]) == false)
            return false;
    return true;
}
```

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Framework- Welcome

- Welcome Page

```
system("Color 0A");

cout << "Welcome to" << endl;

cout << "HHHHHHHHH   HHHHHHHHH                lllllll  iiii" << endl;
cout << "H:::::::H   H:::::::H                l:::::l i::::i" << endl;
cout << "H:::::::H   H:::::::H                l:::::l  iiii" << endl;
cout << "HH::::::H   H::::::HH                l:::::l" << endl;
cout << "H:::::H     H:::::H     eeeeeeeeeeee   l::::l iiiiiiixxxxxxx      xxxxxxx" << endl;
cout << "H:::::H     H:::::H   ee::::::::::::ee l::::l i:::::i x:::::x    x:::::x" << endl;
cout << "H::::::HHHHH::::::H  e::::::eeeee:::::el::::l i::::i  x:::::x  x:::::x" << endl;
cout << "H:::::::::::::::::H e::::::e     e:::::l::::l i::::i   x::::xx:::::x" << endl;
cout << "H:::::::::::::::::H e:::::::eeeee::::::l::::l i::::i    x::::::::::x" << endl;
cout << "H::::::HHHHH::::::H e:::::::::::::::::el::::l i::::i     x::::::::x" << endl;
cout << "H:::::H     H:::::H e::::::eeeeeeeeeee l::::l i::::i     x::::::::x" << endl;
cout << "H:::::H     H:::::H e:::::::e          l::::l i::::i    x::::::::::x" << endl;
cout << "HH::::::H   H::::::H e::::::::e       l::::::i::::::i  x::::xx::::x" << endl;
cout << "H:::::::H   H:::::::H e::::::::eeeeeeeel::::::i::::::i x::::x  x::::x" << endl;
cout << "H:::::::H   H:::::::H ee::::::::::::::el::::::i::::::ix::::x    x::::x" << endl;
cout << "HHHHHHHHH   HHHHHHHHH  eeeeeeeeeeeeeeellllllliiiiiiixxxxxx      xxxxxx" << endl;
```

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Framework-Reading

- Prepare the data from Helix.txt
- Form a String

```
string rand;
//////////////////////////////////////Open File//////////////////////////////////////////////////////
fstream outFile;
outFile.open("Helix.txt");

ifstream HelixV("Helix.txt");

string PassCode;
string UserName;
int counter = 0;
string count;
while (getline(HelixV, PassCode))
{
    if (counter == 0) {
        UserName = PassCode;
        count = PassCode;
    }

    counter++;
}
```

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Framework- Preparation

- Prepare a variable
- T= Number of element in Username

```
int t = 0;
istringstream issCount(count);
string element;
/////Calculate Element number
while (issCount >> element)
{
    t++;
}
```

Framework-Extraction

- Extract data into two Dynamic Array

```
////////Non static array using pointer
string* user = new string[t];
string* password = new string[t];

istringstream iss(UserName);   //extract data into user array
string word;
t = 0;
while (iss >> word) {
    user[t] = word;
    //cout << user[t] << " ";
    t++;
}

istringstream iss2(PassCode);   //extract data into password array
string word2;
t = 0;
//cout << endl;
while (iss2 >> word2) {
    password[t] = word2;
    //cout << password[t] << " ";
    t++;
}

counter = t;
HelixV.close();
```

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Dynamic Array

- Uses Pointer
- Allow the element number of array
- Non Static



Login Selection-User

- Identify User

```
//////////////////////////Login Selection Process
switch (LoginSelection) {
    //User in
case 1: {
    system("Color 0A");
    cout << "Enter your user name" << endl;
    cin >> userin;
    //Eliminate error
    for (int selection = 0; selection <= t; selection++)
    {
        if (user[selection] == userin) {
            LoginSelection = 2;
            CorrectPassword = selection;
            break;
        }

        else if (selection == t) {
            system("CLS");
            system("Color 4F");
            cout << "Error : Wrong Username!!!!!!!!!!!!!!!" << endl;
            cout << "Please Reenter User ID!!!!!!!!!!!!!!!" << endl;
            cout << "Press anything to continue" << endl;
            cin >> rand;
            system("CLS");
            LoginSelection = 1;
            break;

        }
    }
```

ERROR!!!!


Error : Wrong Username!!!!!!!!!!!!!!!
Please Reenter User ID!!!!!!!!!!!!!!!
Press anything to continue

Login Selection-Password

- Use suspend and push
- Replace input as asterisk displayed

```
case 2: {
    int lockscreen;
    string passwordin;
    system("Color 0A");
    cout << "Please enter your password" << endl;;
    lockscreen = _getch();//suspend showing of keyboard until further notice

    while (lockscreen != 13)//convert typing into asterisk
    {
        passwordin.push_back(lockscreen);
        cout << "*";
        lockscreen = _getch();
    }
```

ERROR!!!!



Login Selection- Main Menu

- Menu

```
case 3: {
    int SecLoop = 0;
    string SecSelectionMain;
    system("cls");
    system("Color 0A");
    cout << "Welcome" << endl;
    Sleep(1000);
    cout << "Username: " << "\t" << user[CorrectPassword] << endl;
    Sleep(1000);
    cout << "ToolKit V-1 is here to serve " << endl;
    while (SecLoop == 0) {
        /////Main Menu
        system("Color 0A");
        cout << "1 for Boolean related calculation" << endl;
        cout << "2 for Logic Gate Related issue" << endl;
        cout << "3 for Binary related calculation" << endl;
        cout << "4 for new user and password" << endl;
        cout << "5 for changing password" << endl;
        cout << "   100 for exiting the program" << endl;
        cin >> SecSelectionMain;
```

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Framework-Menu Selection

- Uses Case

```
if (NumChecker(SecSelectionMain)) {
    int SecSelection = stoi(SecSelectionMain);
    if (SecSelection <= 5 && SecSelection > 0 || SecSelection == 100)
    {

        system("CLS");
        switch (SecSelection)

        {

        case 1: { ... }

        case 2: { ... }

        case 3: { ... }

        case 4: { ... }

        case 5: { ... }

        case 100: { ... }
        }

    }
    else {
```

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ERROR!!! AGAIN



Function-Add User/Password

- Add User/Password

```
int comfirmation;
cout << "Enter new User ID" << endl;
cin >> NewUserIn;
cout << " Enter new Password" << endl;
cin >> NewPasswordIn;
cout << "New User ID is " << NewUserIn << "\t" << "New Password is " << NewPasswordIn << endl;
Sleep(1000);
cout << "Enter 1 to comfirm" << endl;
cin >> comfirmation;
```

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Function-Add User/Password

- New Dynamic Array
- Allow one more element to be entered

```
t = t + 1;
system("Color 5E");
string* Nuser = new string[t];
string* Npassword = new string[t];
```

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Function-Add User/Password

- Replace old array with new one

```
ofstream myfile;
myfile.open("Helix.txt");
if (comfirmation == 1)

{
    for (int i = 0; i < t - 1; i++)
    {
        Nuser[i] = user[i];
        Npassword[i] = password[i];
    }

    Nuser[t - 1] = NewUserIn;
    Npassword[t - 1] = NewPasswordIn;

    for (int i = 0; i < t; i++)
    {

        myfile << Nuser[i] << " ";

    }

    myfile << endl;


    for (int i = 0; i < t; i++)
    {

        myfile << Npassword[i] << " ";

    }


}
```

Function-Add User/Password

- Log out to prevent error

```
    }
    myfile.close();
    system("CLS");
    SecLoop = 10000;
    system("CLS");
    cout << "Good Bye " << "\t" << user[CorrectPassword] << endl;
    Sleep(1000);
    cout << "Returning to Login Page" << endl;
    Sleep(1000);
    LoginSelection = 1;
    system("CLS");
    break;
}
```

Function- New Password

- Initialization

```
system("Color 6F");
string OldPass;
string NewPass;


int PassLoopPrimary = 0;
```

Function- New Password

- Check whether Old Password is correct

```
while (PassLoopPrimary == 0) {
    cout << "Enter old password ";
    cin >> OldPass;
    int PassLoopSecondary = 2;
    while (PassLoopSecondary == 2)
    {

        if (OldPass == password[CorrectPassword])
        {
            cout << "Enter new Password" << endl;
            cin >> NewPass;

            cout << "Your new password is " << NewPass << endl;
            cout << "Type 1 to continue" << endl;
            cout << "Type 2 to change password again" << endl;
            cin >> PassLoopSecondary;
            PassLoopPrimary = PassLoopSecondary;


        }
```

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Function- New Password

- Report Error if not

```
        else {
            cout << "Error" << endl;
            cout << "Wrong PAsSwOrd" << endl;
            Sleep(1000);
            system("CLS");
            PassLoopSecondary = 1;
        }
    }
}
```

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Function- New Password

- Write in new password

```
}
//cout << "B4" << endl;
password[CorrectPassword] = NewPass;
//cout << "After" << endl;
ofstream myfile;
myfile.open("Helix.txt");


for (int i = 0; i < t; i++)
{

    myfile << user[i] << " ";

}

myfile << endl;


for (int i = 0; i < t; i++)
{

    myfile << password[i] << " ";

}
//cout << "final" << endl;
```

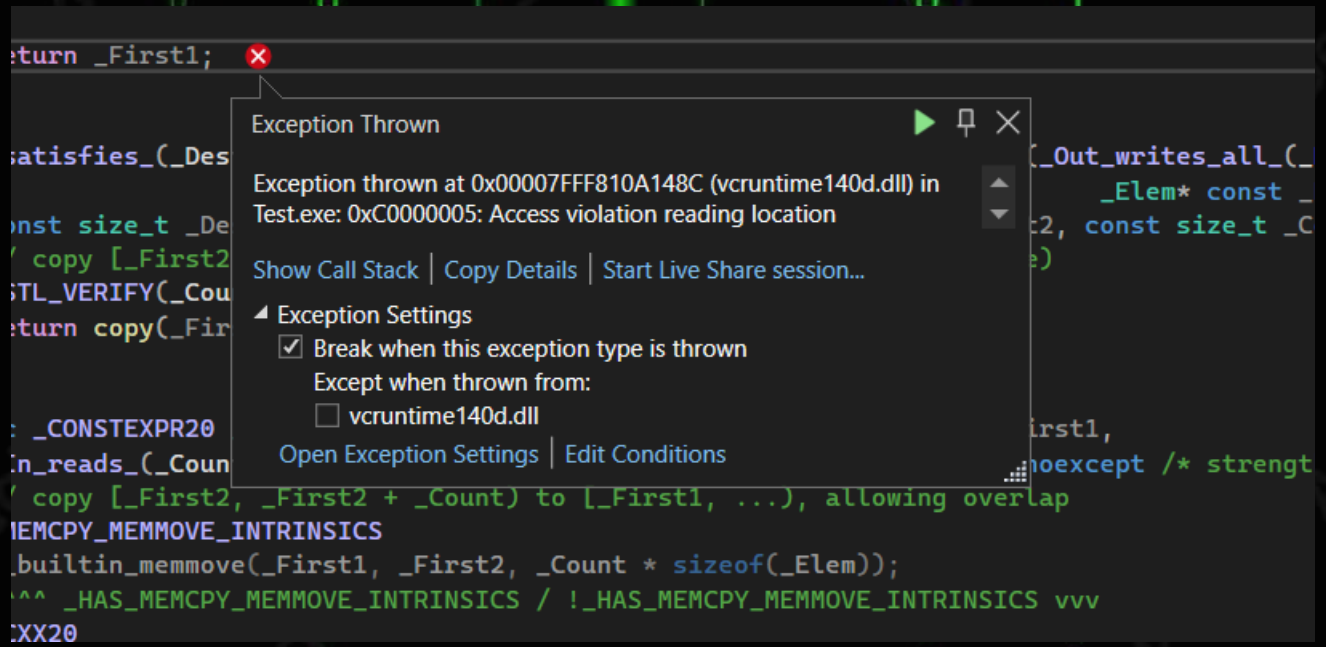
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Function- New Password

- Log out to prevent error

```
    //cout << "final" << endl;
    system("CLS");
    SecLoop = 10000;
    system("CLS");
    cout << "Good Bye " << "\t" << user[CorrectPassword] << endl;
    Sleep(1000);
    cout << "Returning to Login Page" << endl;
    Sleep(1000);
    LoginSelection = 1;
    system("CLS");
    break;
}
```

Error Type


eturn _First1;
Exception Thrown
Exception thrown at 0x00007FFF810A148C (vcruntime140d.dll) in Test.exe: 0xC0000005: Access violation reading location
Show Call Stack | Copy Details | Start Live Share session...
Exception Settings
Break when this exception type is thrown
Except when thrown from:
vcruntime140d.dll
Open Exception Settings | Edit Conditions
satisfies_(_Des
_Out_writes_all_(_
_Elem* const _
nst size_t _De
2, const size_t _C
copy [_First2
e)
STL_VERIFY(_Cou
eturn copy(_Fir
_CONSTEXPR20
irst1,
n_reads_(_Coun
noexcept /* strengt
copy [_First2, _First2 + _Count) to [_First1, ...), allowing overlap
EMCPY_MEMMOVE_INTRINSICS
_builtin_memmove(_First1, _First2, _Count * sizeof(_Elem));
^^ _HAS_MEMCPY_MEMMOVE_INTRINSICS / !_HAS_MEMCPY_MEMMOVE_INTRINSICS vvv
XX20

Function- Log Out

- Log out
- Or Exit Completely

```
case 100: {   int ExitVar = 0;
    system("Color 8F");
    cout << "Sure to Exit? Press 100 to Exit Program" << endl;
    cout << " 1 to Exit to Login Page" << endl;
    cout << " Enter other value to continue " << endl;
    cin >> ExitVar;

    if (ExitVar == 100)
    {
        SecLoop = 10000;
        loop = 10000;
        system("CLS");
        Sleep(1000);
        cout << " Good Bye " << endl;
        cout << " May the beauty of the world always shine " << endl;
        Sleep(1000);
    }
    else if (ExitVar == 1)
    {
        SecLoop = 10000;
        system("CLS");
        cout << "Good Bye " << "\t" << user[CorrectPassword] << endl;
        Sleep(1000);
        cout << "Returning to Login Page" << endl;
        Sleep(1000);
        LoginSelection = 1;
        system("CLS");

    }
```

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Function-Gate Type

- 2 Selection
- Gate/ Truth Table

```
system("Color 3C");
// Initialization
string RawInput_1;
string RawInput_2;
string RawInput_3;
string Gate_TypeRaw;
int slc = 0;
int Input_Loop = 0;
string cont;
int F2Loop = 0;

string NC2S;
cout << "Enter 1 for truth table " << endl;
cout << "Enter 2 for Gate related calculation" << endl;
cin >> NC2S;
```

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Function-Gate Output

- 11 Gates available

```
while (F2Loop == 0) {

    cout << "Key in Input Number" << endl;
    cout << "1 for AND" << endl << "2 for OR" << endl << "3 for NAND" << endl << "4 for NOR" << endl << "5 for Full Add

    cin >> Gate_TypeRaw;
    if (NumChecker(Gate_TypeRaw)) {
        int Gate_Type = stoi(Gate_TypeRaw);
        system("CLS");
        switch (Gate_Type)
```

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Gate-AND

Either input =1

Output =1

```
case 1:
{   cout << "AND GATE" << endl;
cout << "Provide First Input" << endl;
cin >> RawInput_1;
cout << "Provide Second Input" << endl;
cin >> RawInput_2;

if (NumChecker(RawInput_1) && NumChecker(RawInput_2)) { ... }
else { ... }
cout << " Press anything to continue" << endl;
cin >> cont;
system("CLS");

break;
}
```

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Gate-OR Gate

- Either input =1
- Output =1

```
case 2:
{   cout << "OR GATE" << endl;
cout << "Provide First Input" << endl;
cin >> RawInput_1;
cout << "Provide Second Input" << endl;
cin >> RawInput_2;
if (NumChecker(RawInput_1) && NumChecker(RawInput_2)) { ... }
else { ... }
cout << " Press anything to continue" << endl;
cin >> cont;
system("CLS");
break;
}
```

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Gate-NAND

- Inverse AND Gate

```
case 3:
{   cout << "NAND Gate" << endl;
cout << "Provide First Input" << endl;
cin >> RawInput_1;
cout << "Provide Second Input" << endl;
cin >> RawInput_2;
if (NumChecker(RawInput_1) && NumChecker(RawInput_2)) { ... }
else { ... }
cout << " Press anything to continue" << endl;
cin >> cont;
system("CLS");
break;
}
```

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Gate-NOR

- Inverse OR Gate

```
case 4:
{   cout << "NOR GATE" << endl;
cout << "Provide First Input" << endl;
cin >> RawInput_1;
cout << "Provide Second Input" << endl;
cin >> RawInput_2;
if (NumChecker(RawInput_1) && NumChecker(RawInput_2)) { ... }
else { ... }
cout << " Press anything to continue" << endl;
cin >> cont;
system("CLS");
break;
```

Gate-FULL ADDER

- Adding 3 inputs together

```
case 5:
{   cout << "Full Adder" << endl;
cout << "Provide First Input (A) " << endl;
cin >> RawInput_1;
cout << "Provide Second Input (B) " << endl;
cin >> RawInput_2;
cout << "Provide Third Input (C)" << endl;
cin >> RawInput_3;
if (NumChecker(RawInput_1) && NumChecker(RawInput_2) && NumChecker(RawInput_2)) { ... }
else { ... }
cout << " Press anything to continue" << endl;
cin >> cont;
system("CLS");

break;
}
```

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Gate-Half Adder

- Add two inputs together

```
case 6:
{   cout << "Half Adder" << endl;
cout << "Provide First Input" << endl;
cin >> RawInput_1;
cout << "Provide Second Input" << endl;
cin >> RawInput_2;
if (NumChecker(RawInput_1) && NumChecker(RawInput_2)) { ... }
else {
    cout << "Error : Invalid Input" << endl;
    cout << "This gate only accept 1 and 0" << endl;
}
cout << " Press anything to continue" << endl;
cin >> cont;
system("CLS");
break;
}
```

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Gate-Binary Subtractor

- Subtract 3 inputs
- Provide 2 outputs

```
case 7:
{   cout << "Binary Subtractor" << endl;
cout << "Provide First Input (A) " << endl;
cin >> RawInput_1;
cout << "Provide Second Input (B) " << endl;
cin >> RawInput_2;
cout << "Provide Third Input (Borrow-In)" << endl;
cin >> RawInput_3;
if (NumChecker(RawInput_1) && NumChecker(RawInput_2) && NumChecker(RawInput_2)) { ... }
else { ... }
cout << " Press anything to continue" << endl;
cin >> cont;
system("CLS");
break;
}
```

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Gate-SR Flip Flop

- Normal Flip Flop

```
case 8:
{   cout << "SR FLip Flop" << endl;
cout << "Provide First Input S" << endl;
cin >> RawInput_1;
cout << "Provide Second Input R" << endl;
cin >> RawInput_2;
if (NumChecker(RawInput_1) && NumChecker(RawInput_2)) { ... }
else { ... }
cout << " Press anything to continue" << endl;
cin >> cont;
system("CLS");
break;
}
```

Gate-D Flip Flop

- Inverse Input
- D is on

```
case 9:
{   cout << "D Flip Flop" << endl;
cout << "Provide First Input D" << endl;
cin >> RawInput_1;
cout << "Provide current state " << endl;
cin >> RawInput_2;
if (NumChecker(RawInput_1) && NumChecker(RawInput_2)) { ... }
else {
    cout << "Error" << endl;
    cout << "Accept integer only " << endl;
}
cout << " Press anything to continue" << endl;
cin >> cont;
system("CLS");
break;
}
```

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Gate-T Flip Flop

- Inverse Input

```
case 10:
{   cout << "T Flip Flop" << endl;
cout << "Provide First Input t" << endl;
cin >> RawInput_1;
cout << "Provide current state " << endl;
cin >> RawInput_2;
if (NumChecker(RawInput_1) && NumChecker(RawInput_2)) { ... }
else { ... }
cout << " Press anything to continue" << endl;
cin >> cont;
system("CLS");
break;
}
```

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Gate-JK Flip Flop

- Inverse Input Gate with Memory

```
case 11:
{   cout << "JK Flip Flop" << endl;
cout << "Provide First Input J" << endl;
cin >> RawInput_1;
cout << "Provide Second Input K " << endl;
cin >> RawInput_2;
cout << "Provide Current State" << endl;
cin >> RawInput_3;
if (NumChecker(RawInput_1) && NumChecker(RawInput_2) && NumChecker(RawInput_2)) { ... }
else { ... }
cout << " Press anything to continue" << endl;
cin >> cont;
system("CLS");
break;
}
```

Function- Return to Main Menu

- Inverse Input with Memory

```
case 12:
{cout << "Returning to Main Menu" << endl;
Sleep(1000);
system("CLS");
F2Loop = 1000;
break;
}
```

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Function- Error Display

```
    default: {
        cout << "Error" << endl;
        cout << "Enter a value from 1 to 12" << endl;
        Sleep(1000);
        system("CLS");
    }

}
```

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Binary Related Issues

```
Welcome to binary related issue page!

1. Parity system test simulator
2. 8-bit signed binary numbers' systems converter

Please select the function[1/2]:
```

```cpp
int main()
{
    string selection;
    string decide;

    cout << "Welcome to binary related issue page!\n" << endl;
    cout << "1. Parity system test simulator" << endl;
    cout << "2. 8-bit signed binary numbers' systems converter\n" << endl;

    do
    {
        do
        {
            cout << "Please select the function[1/2]: ";
            cin >> selection;
```

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```
Welcome to binary related issue page!

1. Parity system test simulator
2. 8-bit signed binary numbers' systems converter

Please select the function[1/2]: 123
Invalid Input!
Please select the function[1/2]: a1d
Invalid Input!
Please select the function[1/2]:
```

```
do
{
    cout << "Please select the function[1/2]: ";
    cin >> selection;
    if (selection != "1" && selection != "2")
        cout << "Invalid Input!";
    cout << endl;
} while (selection != "1" && selection != "2");
```

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```
Welcome to binary related issue page!

1. Parity system test simulator
2. 8-bit signed binary numbers' systems converter

Please select the function[1/2]: 123
Invalid Input!
Please select the function[1/2]: a1d
Invalid Input!
Please select the function[1/2]: 1

Welcome to parity system test simulator!

Please enter parity system used by your electronic device [even/odd]:
```

```
if (selection == "1")
{
    func1();
}
```

```
void func1()
{
    string code;
    string new_code;
    string parity;
    string status;
    string decision;
    string error;
    int length;
    int length_parity;
    int length_decision;

    cout << "Welcome to parity system test simulator!\n\n";

    do
    {
        do
        {
            cout << "Please enter parity system used by your electronic device [even/odd]: "; // prompt the guest to enter the parity system used
            cin >> parity; // enter parity system used
            length_parity = parity.size();
            for (int j = 0; j < length_parity; j++) //To change each element in parity to lowercase
            {
                parity[j] = tolower(parity[j]);
            }
```

```
Welcome to parity system test simulator!

Please enter parity system used by your electronic device [even/odd]: 123
Invalid input!
Please enter parity system used by your electronic device [even/odd]: dws1
Invalid input!
Please enter parity system used by your electronic device [even/odd]:
```

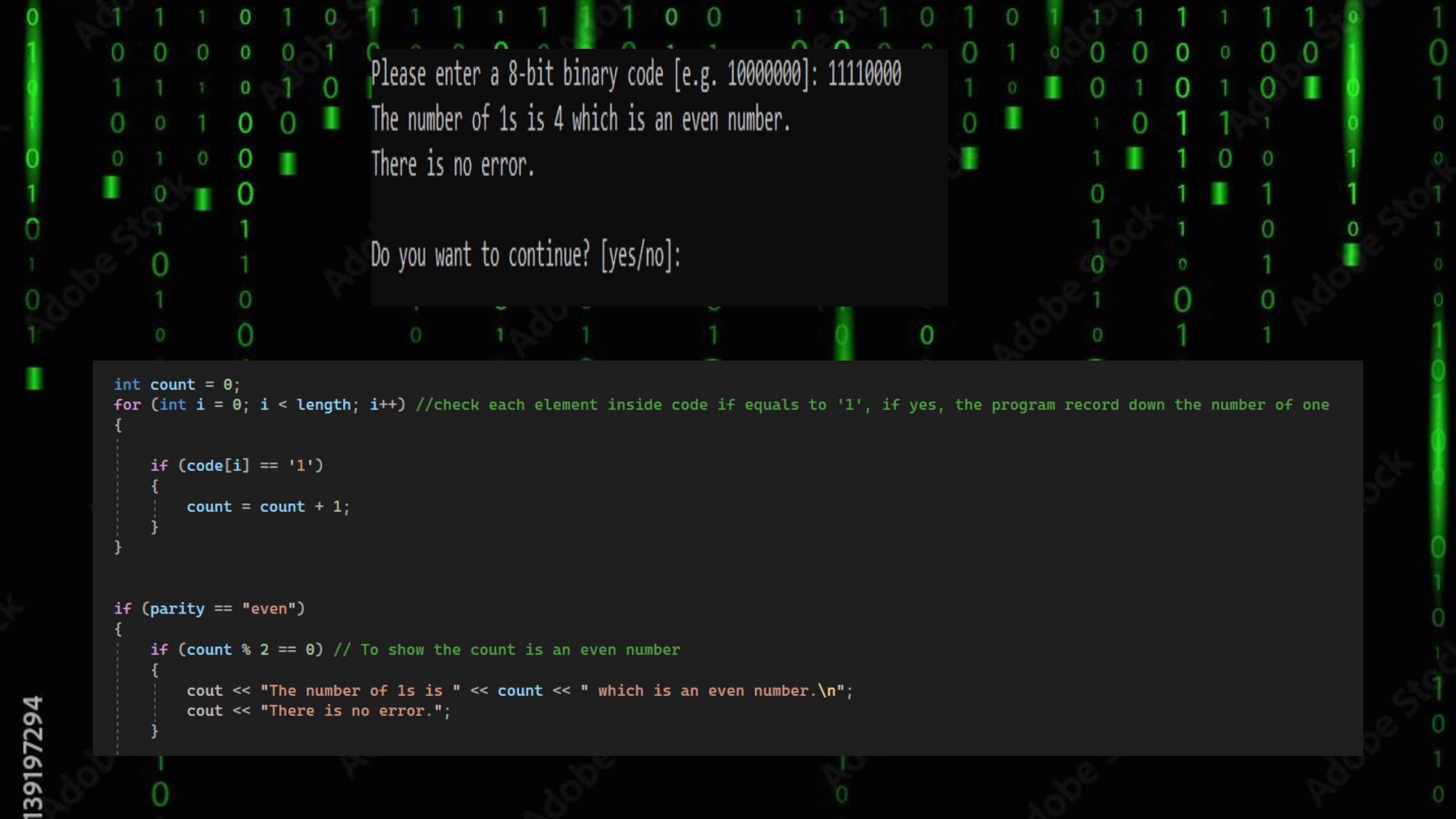
```cpp
do
{
    cout << "Please enter parity system used by your electronic device [even/odd]: "; // prompt the guest to enter the parity system used
    cin >> parity; // enter parity system used
    length_parity = parity.size();
    for (int j = 0; j < length_parity; j++) //To change each element in parity to lowercase
    {
        parity[j] = tolower(parity[j]);
    }
    if (parity != "even" && parity != "odd")
        cout << "Invalid input!\n";
} while (parity != "even" && parity != "odd"); //Ask the guest enter again if input is wrong
```

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```
Please enter parity system used by your electronic device [even/odd]: even
Please enter a 8-bit binary code [e.g. 10000000]: 1111111
Invalid input!
Please enter a 8-bit binary code [e.g. 10000000]: 111111111
Invalid input!
Please enter a 8-bit binary code [e.g. 10000000]: dfgd12
Invalid input!
Please enter a 8-bit binary code [e.g. 10000000]:
```

```cpp
do
{
    cout << "Please enter a 8-bit binary code [e.g. 10000000]: "; //prompt the guest to enter 8-bit binary code
    cin >> code; // enter the code
    length = code.size(); //To get the size or length of code
    for (int k = 0; k < length; k++) //if the element in the code is not 1 and 0, there is an error
    {                                //if the binary code entered is not 8-bit, there is an error
        if ((code[k] != '1' && code[k] != '0') || length != 8)
        {
            error = "yes";
            cout << "Invalid input!\n";
            break;
        }
        else
            error = "no";
    }
} while (error == "yes"); //if error == yes, means there is invalid input, the program will ask the guest to enter again
```

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```
int count = 0;
for (int i = 0; i < length; i++) //check each element inside code if equals to '1', if yes, the program record down the number of one
{

    if (code[i] == '1')
    {
        count = count + 1;
    }
}


if (parity == "even")
{
    if (count % 2 == 0) // To show the count is an even number
    {
        cout << "The number of 1s is " << count << " which is an even number.\n";
        cout << "There is no error.";
    }
```

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```
Do you want to continue? [yes/no]: yes
Please enter parity system used by your electronic device [even/odd]: even
Please enter a 8-bit binary code [e.g. 10000000]: 11100000
The number of 1s is 3 which is an odd number.
There is an error.
Please enter again [e.g. 10000000]:11000000
The number of 1s is 2 which is an even number.
There is no error.

Do you want to continue? [yes/no]: no


Do you want to select again[yes/no]:
```

```
    cout << "\n\nDo you want to continue? [yes/no]: "; //Ask the guest if want to continue
    cin >> decision;
    length_decision = decision.size();
    for (int j = 0; j < length_decision; j++) //To change each element in decision to lowercase
    {
        decision[j] = tolower(decision[j]);
    }
} while (decision == "yes"); // The program will run again if the decision is yes.
```

```
else
{
    cout << "The number of 1s is " << count << " which is an odd number.\n";
    cout << "There is an error."; //error occurs
    cout << "\n";

    do
    {
        do
        {
            cout << "Please enter again [e.g. 10000000]:"; //ask the guest to enter again
            cin >> new_code;
            length = new_code.size();
            for (int k = 0; k < length; k++) //if the element in the new_code is not 1 and 0, there is an error
            {                                //if the binary code entered is not 8-bit, there is an error
                if ((new_code[k] != '1' && new_code[k] != '0') || length != 8)
                {
                    error = "yes";
                    cout << "Invalid input!\n";
                    break;
                }
                else
                    error = "no";
            }
        } while (error == "yes"); //if error == yes, means there is invalid input, the program will ask the guest to enter again
```

```
Do you want to select again[yes/no]: yes
Please select the function[1/2]: 2

Welcome to 8-bit signed binary numbers' systems converter ^_^

Please enter a decimal number [0 <= magnitude <= 127]:
```

```cpp
do
{
    do
    {
        cout << "Please select the function[1/2]: ";
        cin >> selection;
        if (selection != "1" && selection != "2")
            cout << "Invalid Input!";
        cout << endl;
    } while (selection != "1" && selection != "2");

    if (selection == "1")
    {
        func1();
    }

    else if (selection == "2")
    {
        func2();
    }

    cout << "\n\nDo you want to select again[yes/no]: ";
    cin >> decide;
} while (decide == "Yes" || decide == "YES" || decide == "yes");
```

```cpp
void func2()

    string decision1;
    string decimalnumstr;
    string digit;
    int remainder;
    int decimalnum;
    int array[8]; //set array's size to 8 for 8-bit binary number
    int status1;
    int length_decision1;

    cout << "Welcome to 8-bit signed binary numbers' systems converter ^_^\n";

    do
    {
        int i = 0;
        do
        {
            cout << "\nPlease enter a decimal number [0 <= magnitude <= 127]: "; // ask guest to enter a decimal number
            cin >> decimalnumstr; // enter decimal number in string data type
```

```
do
{
    cout << "\nPlease enter a decimal number [0 <= magnitude <= 127]: "; // ask guest to enter a decimal number
    cin >> decimalnumstr; // enter decimal number in string data type

    digit = checkdigit(decimalnumstr); //call function of checkdigit to check if each character in decimalnumstr
    // is digit. If all the characters are digit, the function returns "true"
    // if got one character is not digit, the function returns "false"
    if (digit == "true")
    {
        decimalnum = stoi(decimalnumstr); //Convert string decimalnumstr to int decimalnum by stoi (use <string> header file)

        if (decimalnum <= 127 && decimalnum >= -127) //if decimalnum is between -127 to 127, the status is 1
            status1 = 1;
        else                                    //else the status is 0 and Invalid input! is printed out.
        {
            status1 = 0;
            cout << "Invalid input!";
        }
    }

    else //else the status is 0 and Invalid input! is printed out.
    {
        status1 = 0;
        cout << "Invalid input!";
    }

} while (status1 == 0); //if decimal number is not between 127 and -127 or the input is not digit, the program ask the guest to enter the decimal value again
```

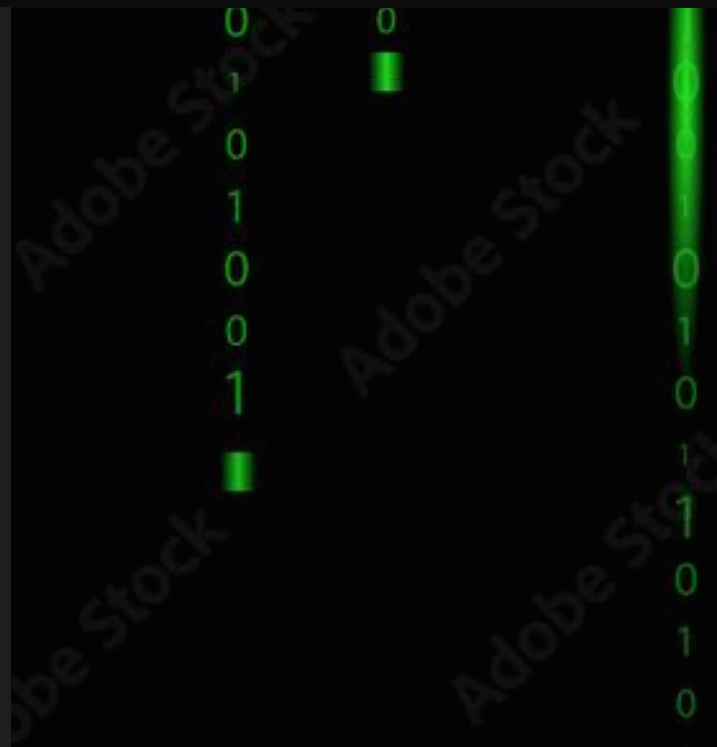
```
Welcome to 8-bit signed binary numbers' systems converter ^_^

Please enter a decimal number [0 <= magnitude <= 127]: 128
Invalid input!
Please enter a decimal number [0 <= magnitude <= 127]: 1df
Invalid input!
Please enter a decimal number [0 <= magnitude <= 127]:
```

```
string checkdigit(string decimalnumstr)
{
    int length_decimalnumstr = decimalnumstr.size(); //To get size or length of decimalnumstr.
    string status;

    if (decimalnumstr[0] == '-') //if the first character is '-', the program will ignore it and check from next character.
    {
        for (int i = 1; i < length_decimalnumstr; i++) //check from the character after '-'.
        {
            if (!isdigit(decimalnumstr[i])) //if the charater is not a digit, the status = "false" and the for loop break.
            {
                status = "false";
                break;
            }
            else
                status = "true"; //if the character is a digit, the status = "true"
        }
    }

    else
    {
        for (int i = 0; i < length_decimalnumstr; i++) //check from first value to the end.
        {
            if (!isdigit(decimalnumstr[i])) //if the charater is not a digit, the status = "false" and the for loop break.
            {
                status = "false";
                break;
            }
            else
                status = "true"; //if the character is a digit, the status = "true".
        }
    }
    return status; //return status either "true" or "false".
}
```



```
Please enter a decimal number [0 <= magnitude <= 127]: 1

Sign and magnitude system: 00000001
1's complement system: 00000001
2's complement system: 00000001

Do you want to continue? [yes/no]:yes

Please enter a decimal number [0 <= magnitude <= 127]: -1

Sign and magnitude system: 10000001
1's complement system: 11111110
2's complement system: 11111111

Do you want to continue? [yes/no]:
```

```
if (decimalnum >= 0 && decimalnum <= 127) // for positive decimal number case
{
    while (decimalnum != 0)  // when decimalnum == 0, no more division is required
    {

        remainder = decimalnum % 2; // To calculate remainder by % 2 because of binary system
        array[i++] = remainder;  // current remainder value store in array[i]
        decimalnum = decimalnum / 2;  // i = i + 1 , To calculate new decimalnum

    }

    positivenum(i, array); // To convert positive decimal number to signed binary numbers' systems


}
```

```
void positivenum(int i, int array[])
{
    while (i < 8)  // when number of bit less than 8, add zero until 8 bit
    {
        array[i++] = 0;
    }

    cout << "Sign and magnitude system: ";
    for (int j = 7; j >= 0; j--) //print out the binary number in an correct order by starting from the highest index of the array
    {
        cout << array[j];
    }

    cout << endl;

    cout << "1's complement system: ";
    for (int j = 7; j >= 0; j--) //print out the binary number in an correct order by starting from the highest index of the array
    {
        cout << array[j];
    }

    cout << endl;

    cout << "2's complement system: ";
    for (int j = 7; j >= 0; j--) //print out the binary number in an correct order by starting from the highest index of the array
    {
        cout << array[j];
    }
}
```

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```
Please enter a decimal number [0 <= magnitude <= 127]: 1

Sign and magnitude system: 00000001
1's complement system: 00000001
2's complement system: 00000001

Do you want to continue? [yes/no]:yes

Please enter a decimal number [0 <= magnitude <= 127]: -1

Sign and magnitude system: 10000001
1's complement system: 11111110
2's complement system: 11111111

Do you want to continue? [yes/no]:
```

```
else if (decimalnum <= -1 && decimalnum >= -127) //for negative decimal number case
{
    decimalnum = -1 * decimalnum; //to get the magnitude of decimal number

    while (decimalnum != 0)  // when decimalnum == 0, no more division is required
    {

        remainder = decimalnum % 2; // To calculate remainder by % 2 because of binary system
        array[i++] = remainder;  // current remainder value store in array[i]
        decimalnum = decimalnum / 2;  // i = i + 1 , To calculate new decimalnum

    }

    signmagnitude(i, array); //Convert to sign and magnitude system
    cout << endl;

    firstcom(i, array); //Convert to 1's complement system
    cout << endl;

    secondcom(i, array); //Convert to 2's complement system
}
```

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```
Do you want to continue? [yes/no]:no


Do you want to select again[yes/no]: no
Good bye!
```

```
  cout << "\n\nDo you want to continue? [yes/no]:";
  cin >> decision1;
  length_decision1 = decision1.size();
  for (int j = 0; j < length_decision1; j++) //To change each element in decision to lowercase
  {
      decision1[j] = tolower(decision1[j]);
  }
while (decision1 == "yes"); //if the guest enter yes as decision, the program will run again
```

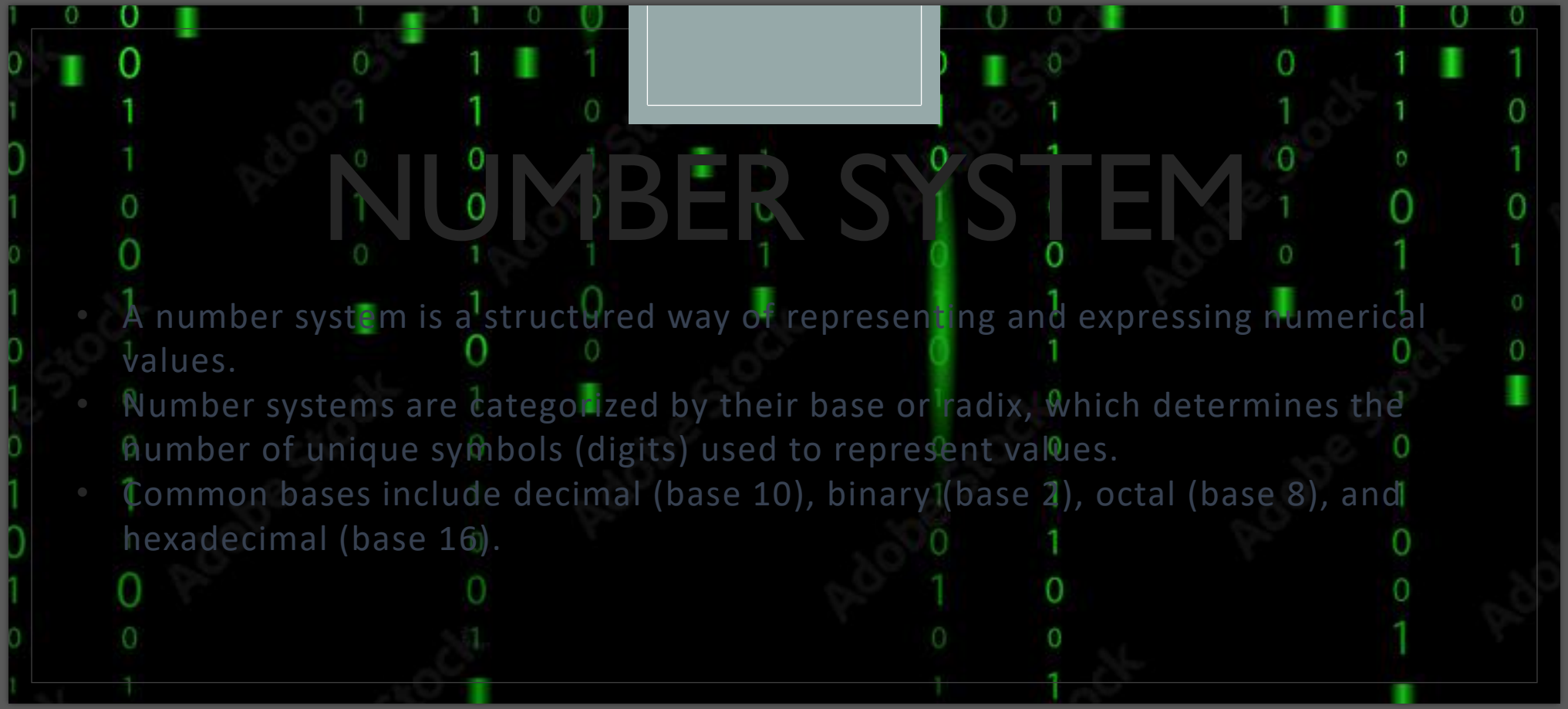
```
    cout << "\n\nDo you want to select again[yes/no]: ";
    cin >> decide;
  } while (decide == "Yes" || decide == "YES" || decide == "yes");

  cout << "Good bye!";
  return 0;
```

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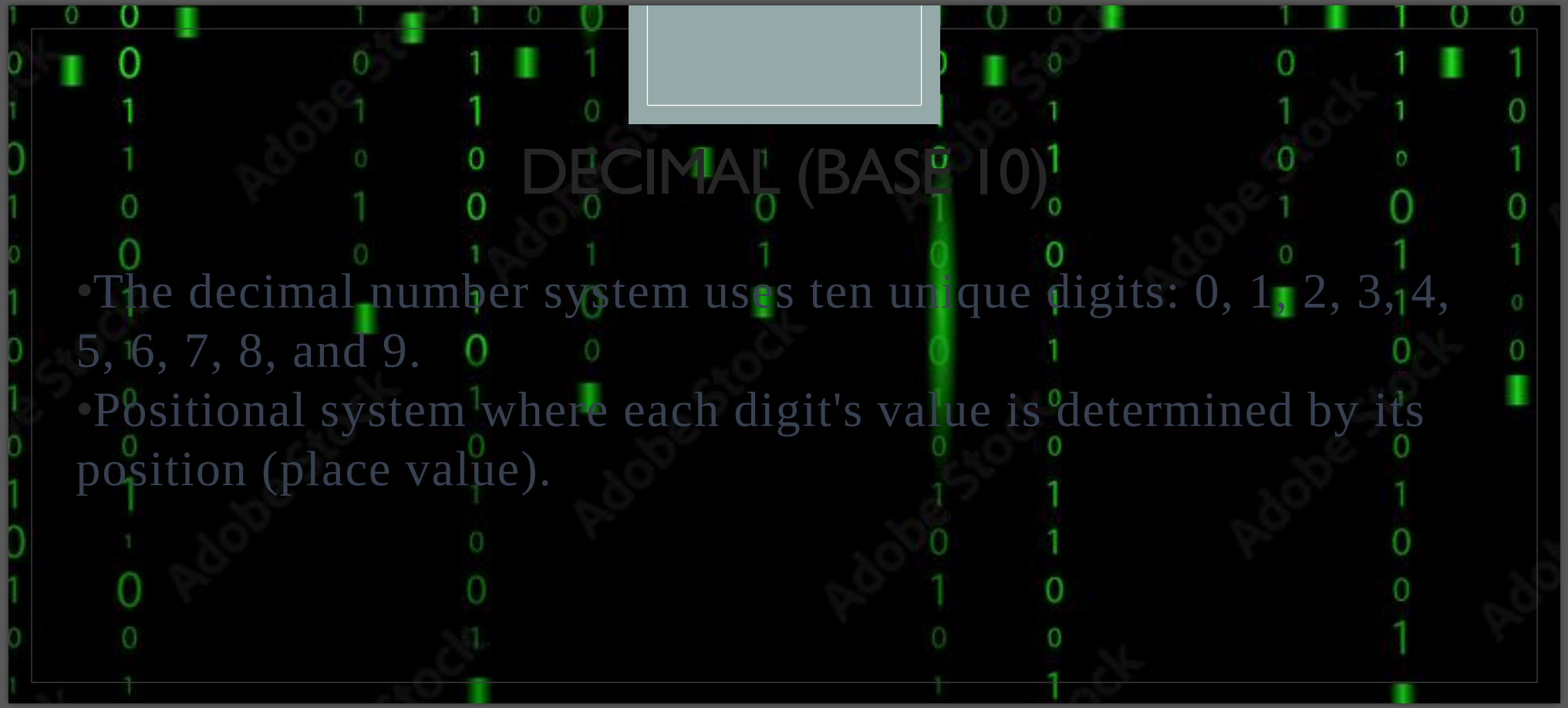
NUMBER SYSTEM

Presented by:
Poh Zheng Yang



NUMBER SYSTEM

- A number system is a structured way of representing and expressing numerical values.
- Number systems are categorized by their base or radix, which determines the number of unique symbols (digits) used to represent values.
- Common bases include decimal (base 10), binary (base 2), octal (base 8), and hexadecimal (base 16).



DECIMAL (BASE 10)

- The decimal number system uses ten unique digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9.
- Positional system where each digit's value is determined by its position (place value).

BINARY (BASE 2)

- The binary number system uses only two digits: 0 and 1.
- Widely used in computing due to its compatibility with electronic systems.

Example:

2(Decimal) = 10(Binary)
6(Decimal) = 110(Binary)
69(Decimal) = 100101(Binary)

OCTAL (BASE 8)

- The octal number system uses eight unique digits: 0 to 7.
- Rarely used today but was more common in older computer systems.

Example:

8(Decimal) = 10(Octal)

89(Decimal) = 131(Octal)

1380(Decimal) = 2544(Octal)

HEXADECIMAL (BASE 16)

- The hexadecimal number system uses sixteen unique digits: 0-9 and A-F, where A stands for 10, B for 11, and so on.
- Frequently used in programming for concise representation of binary data.

Example:

16(Decimal) = 10(Hexadecimal)

10(Decimal) = A(Hexadecimal)

2585(Decimal) = A19(Hexadecimal)


NUMBER SYSTEM
CONVERSION
Presented by Poh Zheng Yang



BINARY, OCTAL & HEXADECIMAL TO DECIMAL



BINARY TO DECIMAL

Multiplication method

Binary : Determine the positional value of every digit of binary number. Multiply every digit by 2^0 2^1 2^2 2^3 2^4 and so one from right to left.

Example:



OCTAL TO DECIMAL

Multiplication method

Octal : Determine the positional value of every digit of octal number. Multiply every digit by 8^0, 8^1, 8^2, 8^3, 8^4 and so one from right to left.

Example:

HEXADECIMAL TO DECIMAL

Multiplication method

Hexadecimal : Determine the positional value of every digit of hexadecimal number. Multiply every digit by 16^0 16^1 16^2 16^3 16^4 and so one from right to left.

Example:

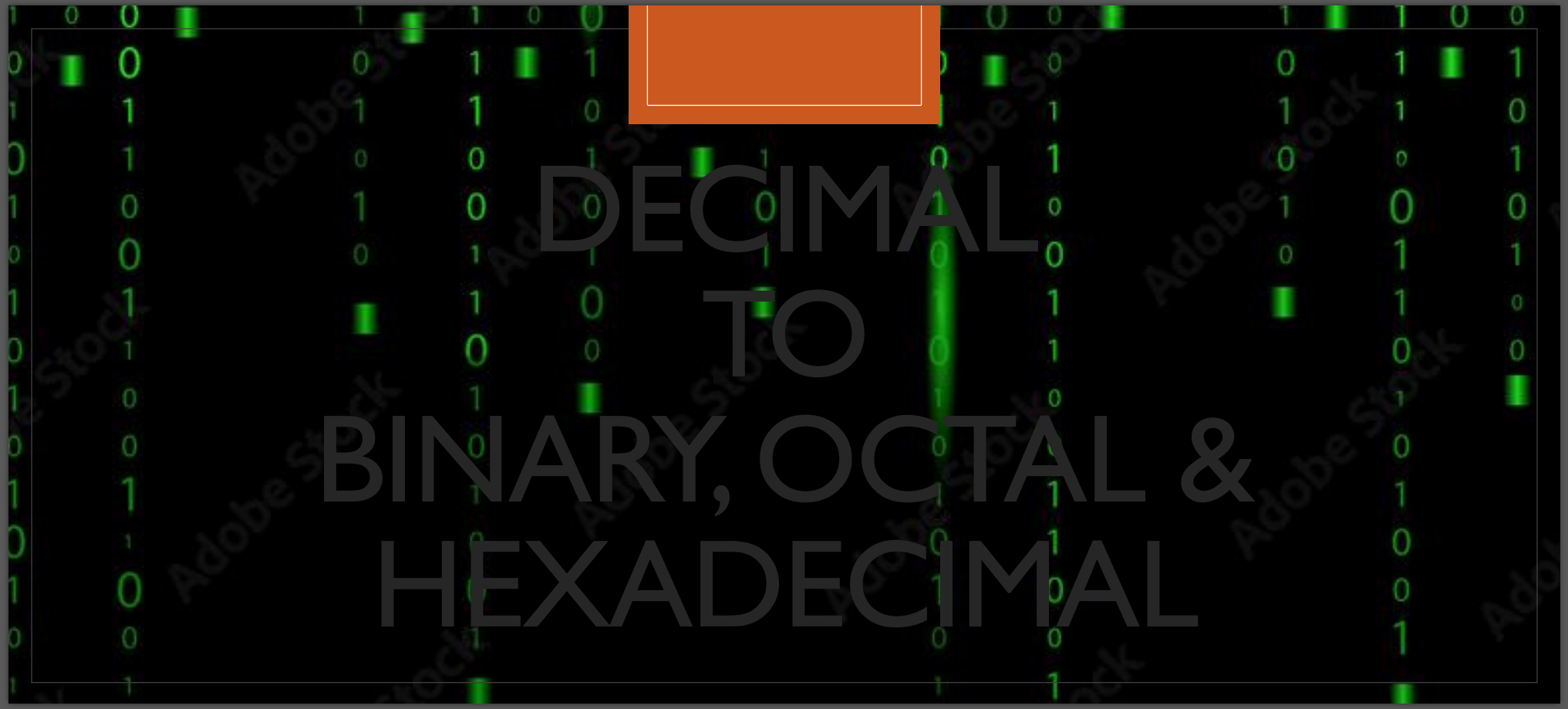
```cpp
// Function to convert binary to decimal
int binaryToDecimal(const string& binary)
{
    int decimal = 0;
    int power = 0;
    for (int i = binary.length() - 1; i >= 0; --i)
    {
        if (binary[i] == '1')
            decimal += pow(2, power);
        power++;
    }
    return decimal;
}
```

- Use For loop to add up all the element inside the array to convert to decimal in integer form

```cpp
// Function to convert hexadecimal to decimal
int hexadecimalToDecimal(const string& hexadecimal)
{
    int decimal = 0;
    int power = 0;
    for (int i = hexadecimal.length() - 1; i >= 0; --i)
    {
        if (hexadecimal[i] >= '0' && hexadecimal[i] <= '9')
            decimal += (hexadecimal[i] - '0') * pow(16, power);
        else if (hexadecimal[i] >= 'A' && hexadecimal[i] <= 'F')
            decimal += (hexadecimal[i] - 'A' + 10) * pow(16, power
        power++;
    }
    return decimal;
}
```

```cpp
// Function to convert octal to decimal
int octalToDecimal(const string& octal)
{
    int decimal = 0;
    int power = 0;
    for (int i = octal.length() - 1; i >= 0; --i)
    {
        decimal += (octal[i] - '0') * pow(8, power);
        power++;
    }
    return decimal;
}
```

Both function (Hexadecimal to decimal) (Octal to decimal) are using same ways for converting to decimal.


DECIMAL
TO
BINARY, OCTAL &
HEXADECIMAL

DECIMAL TO BINARY

Remainder method

Binary: Continue dividing the number by 2 and take the remainder in opposite order which will be the binary equivalent of given decimal number.

Example:

Division	Result	Remainder
27/2	13	1
13/2	6	1
6/2	3	0
3/2	1	1
1/2	0	1

$(27)_{10} = (11011)_2$

DECIMAL TO OCTAL

Remainder method

Binary: Continue dividing the number by 8 and take the remainder in opposite order which will be the octal equivalent of given decimal number.

Example:

Division	Result	Remainder
158/8	19	6
19/8	2	3
2/8	0	2

$(158)_{10} = (236)_8$

DECIMAL TO HEXADECIMAL

Remainder method

Binary: Continue dividing the number by 16 and take the remainder in opposite order which will be the binary equivalent of given decimal number.

Example:

Division	Result	Remainder
450/16	28	2
28/16	1	12(C)
1/16	0	1

$(450)_{10} = (1C2)_{16}$

```
// Function to convert decimal to binary
string decimalToBinary(int decimal)
{
    string binary = "";
    while (decimal > 0)
    {
        binary = to_string(decimal % 2) + binary;
        decimal /= 2;
    }
    return binary;
}
```

- Use For loop to divide the decimal by 2 and add up all the raminders into a string to form an array.

```cpp
// Function to convert decimal to octal
string decimalToOctal(int decimal)
{
    string octal = "";
    while (decimal > 0)
    {
        octal = to_string(decimal % 8) + octal;
        decimal /= 8;
    }
    return octal;
}


// Function to convert decimal to hexadecimal
string decimalToHexadecimal(int decimal)
{
    string hexadecimal = "";
    while (decimal > 0)
    {
        int remainder = decimal % 16;
        if (remainder < 10)
            hexadecimal = to_string(remainder) + hexadecimal;
        else
            hexadecimal = char('A' + remainder - 10) + hexadecimal;
        decimal /= 16;
    }
    return hexadecimal;
}
```

Both function (Decimal to hexadecimal) (Decimal to octal) are using same ways for converting to decimal.

Check input

```cpp
// Function to check binary input
bool binCheck(string number)
{
    for (char bin : number)
    {
        if (bin != '0' && bin != '1')
        {
            return false;
        }
    }
    return true;
}
```

```cpp
// Function to check octal input
bool octalCheck(string number)
{
    for (char oct : number)
    {
        if (oct < '0' || oct > '7')
            return false;
    }
    return true;
}
```

```cpp
// Function to check hexadecimal input
bool hexaCheck(string number)
{
    for (char hex : number)
    {
        if (!((hex >= '0' && hex <= '9') || (hex >= 'A' && hex <= 'F') || (hex >= 'a' && hex <= 'f')))
        {
            return false;
        }
    }
    return true;
}
```

Main function

```cpp
int main()
{
    string number, outputNum;
    int inputType, outputType;
    int decimalNumber;
    bool continueConvert = true;

    while (continueConvert)
    {
        cout << "***********************************************" << endl;
        cout << "Please choose Input type:\n1.Binary\n2.Octal\n3.Decimal\n4.Hexadecimal" << endl;
        cout << "***********************************************" << endl;
        cout << "Input type (1-4): ";
        string inputTypeStr;
        cin >> inputTypeStr;
        cout << "***********************************************" << endl;
        while (!autoCheck(inputTypeStr) || (stoi(inputTypeStr) < 1 || stoi(inputTypeStr) > 4))
        {
            cout << "Invalid Input type! Please re-enter valid Input type." << endl;
            cout << "Input type (1-4): ";
            cin.clear();
            cin.ignore(numeric_limits<streamsize>::max(), '\n');
            cin >> inputTypeStr;
        }
        inputType = stoi(inputTypeStr);
```

Main function (cont)

```
cout << "Please enter number you would like to convert: ";
cin >> number;
cout << "**********************************************" << endl;
if (inputType == 1)
{
    while (!binCheck(number))
    {
        cout << "Invalid input of Binary number! Binary number should only contain 0 and 1 only!" << endl;
        cout << "Please re-enter valid Binary number." << endl;
        cout << "Please enter number you would like to convert: ";
        cin >> number;
    }
    decimalNumber = binaryToDecimal((number));
}
else if (inputType == 2)
{
    while (!octalCheck(number))
    {
        cout << "Invalid input of Octal number! Octal number should only contain integer from 0 - 7 only!" << endl;
        cout << "Please re-enter valid Octal number." << endl;
        cout << "Please enter number you would like to convert: ";
        cin >> number;
    }
    decimalNumber = octalToDecimal((number));
}
```

Main function (cont)

```
else if (inputType == 3)
{
    while (!autoCheck(number))
    {
        cout << "Invalid input of Decimal number! Please enter a valid Decimal number: ";
        cin >> number;
    }
    decimalNumber = stoi(number);
}
else if (inputType == 4)
{
    while (!hexaCheck(number))
    {
        cout << "Invalid input of Hexadecimal number! Hexadecimal number should only contain integer from 0 - 9 and alphabet from
        cout << "Please re-enter valid Hexadecimal number." << endl;
        cout << "Please enter number you would like to convert: ";
        cin >> number;
    }
    decimalNumber = hexadecimalToDecimal((number));
}
else
{
    cout << "Invalid input type. Exiting the program." << endl;
    return 0;
}
```

Main function (cont)

```
cout << "Please choose Output type:\n1.Binary\n2.Octal\n3.Decimal\n4.Hexadecimal" << endl;
cout << "Output type (1-4): ";
string outputTypeStr;
cin >> outputTypeStr;
cout << "************************************************" << endl;
while (!autoCheck(outputTypeStr) || (stoi(outputTypeStr) < 1 || stoi(outputTypeStr) > 4))
{
    cout << "Invalid Output type! Please re-enter valid Output type." << endl;
    cout << "Output type (1-4): ";
    cin.clear();
    cin.ignore(numeric_limits<streamsize>::max(), '\n');
    cin >> outputTypeStr;
}
outputType = stoi(outputTypeStr);

if (outputType == 1)
{
    outputNum = decimalToBinary(decimalNumber);
}
else if (outputType == 2)
{
    outputNum = decimalToOctal(decimalNumber);
}
else if (outputType == 3)
{
    outputNum = to_string(decimalNumber);
```

Main function (cont)

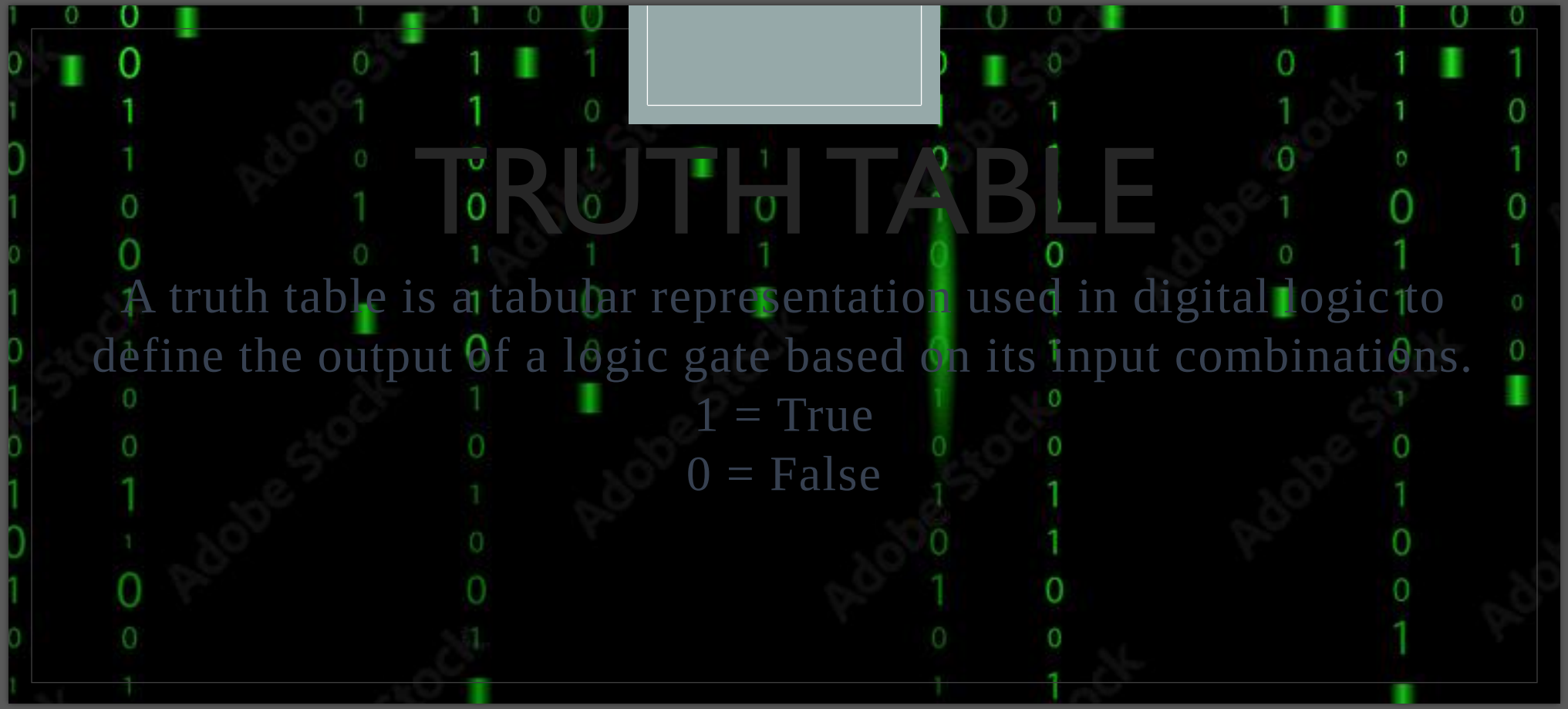
```
    else if (outputType == 4)
    {
        outputNum = decimalToHexadecimal(decimalNumber);
    }
    else
    {
        cout << "Invalid output type. Exiting the program." << endl;
        return 0;
    }

    cout << "Converted number: " << outputNum << endl;
    cout << "**********************************************" << endl;
    char continueConvertChoice;
    cout << "Do you want to continue? (Y/N): ";
    cin >> continueConvertChoice;
    if (toupper(continueConvertChoice) != 'Y')
    {
        continueConvert = false;
    }
}
cout << "**********************************************" << endl;
cout << "Exiting the program." << endl;
return 0;
```



TRUTH TABLE FOR LOGIC GATES

Presented by Poh Zheng Yang



TRUTH TABLE

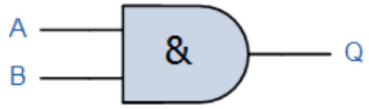
A truth table is a tabular representation used in digital logic to define the output of a logic gate based on its input combinations.

1 = True

0 = False

AND GATE

AND gate: The output Q is true if ALL input are true

Symbol	Truth Table		
 2-input AND Gate	A	B	Q
	0	0	0
	0	1	0
	1	0	0
	1	1	1

Function to perform logical AND

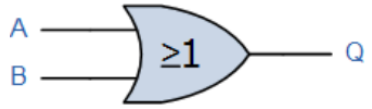
```cpp
// Function to perform logical AND operation
bool logicalAND(const vector<bool>& inputs)
{
    bool result = true;
    for (bool input : inputs)
    {
        result &= input;
    }
    return result;
}
```

Output

```
**********************************************
Select the gate:
1. AND
2. OR
3. NOT
4. XOR
5. XNOR
6. NAND
7. NOR
**********************************************
Enter the number corresponding to the gate: 1
**********************************************
Enter the number of inputs: 2
Truth Table:
0 0 | 0
1 0 | 0
0 1 | 0
1 1 | 1
**********************************************
Do you want to continue? (Y/N):
```

OR GATE

OR gate: The output Q is true if EITHER one of the inputs are true.

Symbol	Truth Table		
 2-input OR Gate	A	B	Q
	0	0	0
	0	1	1
	1	0	1
	1	1	1

Function to perform logical OR

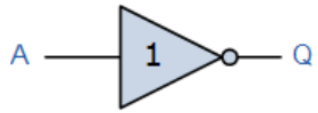
```cpp
// Function to perform logical OR operation
bool logicalOR(const vector<bool>& inputs)
{
    bool result = false;
    for (bool input : inputs)
    {
        result |= input;
    }
    return result;
}
```

Output

```
1. AND
2. OR
3. NOT
4. XOR
5. XNOR
6. NAND
7. NOR
**********************************************
Enter the number corresponding to the gate: 2
**********************************************
Enter the number of inputs: 4
Truth Table:
0 0 0 0 | 0
1 0 0 0 | 1
0 1 0 0 | 1
1 1 0 0 | 1
0 0 1 0 | 1
1 0 1 0 | 1
0 1 1 0 | 1
1 1 1 0 | 1
0 0 0 1 | 1
1 0 0 1 | 1
0 1 0 1 | 1
1 1 0 1 | 1
0 0 1 1 | 1
1 0 1 1 | 1
0 1 1 1 | 1
1 1 1 1 | 1
**********************************************
Do you want to continue? (Y/N):
```

NOT GATE

NOT gate: The output Q is true if the input is false

Symbol	Truth Table	
 Inverter or NOT Gate	A	Q
	0	1
	1	0

Function to perform logical NOT

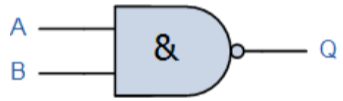
```cpp
// Function to perform logical NOT operation
bool logicalNOT(bool input)
{
    return !input;
}
```

Output

```
**********************************************
Select the gate:
1. AND
2. OR
3. NOT
4. XOR
5. XNOR
6. NAND
7. NOR
**********************************************
Enter the number corresponding to the gate: 3
**********************************************
Enter the number of inputs: 1
Truth Table:
0 | 1
1 | 0
**********************************************
```

NAND GATE

NAND gate: The output Q is false if ALL the input are true. Complete Opposite output as AND gate.

Symbol	Truth Table		
 2-input NAND Gate	A	B	Q
	0	0	1
	0	1	1
	1	0	1
	1	1	0

Function to perform logical NAND

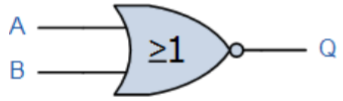
```cpp
// Function to perform logical NAND operation
bool logicalNAND(const vector<bool>& inputs)
{
    bool result = true;
    for (bool input : inputs)
    {
        result &= input;
    }
    return !result;
}
```

Output

```
1. AND
2. OR
3. NOT
4. XOR
5. XNOR
6. NAND
7. NOR
*****************************************
Enter the number corresponding to the gate: 6
*****************************************
Enter the number of inputs: 4
Truth Table:
0 0 0 0 | 1
1 0 0 0 | 1
0 1 0 0 | 1
1 1 0 0 | 1
0 0 1 0 | 1
1 0 1 0 | 1
0 1 1 0 | 1
1 1 1 0 | 1
0 0 0 1 | 1
1 0 0 1 | 1
0 1 0 1 | 1
1 1 0 1 | 1
0 0 1 1 | 1
1 0 1 1 | 1
0 1 1 1 | 1
1 1 1 1 | 0
*****************************************
```

NOR GATE

NOR gate: The output Q is false if EITHER one of the inputs are true. Complete opposite output as OR gate

Symbol	Truth Table		
 2-input NOR Gate	A	B	Q
	0	0	1
	0	1	0
	1	0	0
	1	1	0

Function to perform logical NOR

```cpp
// Function to perform logical NOR operation
bool logicalNOR(const vector<bool>& inputs)
{
    bool result = false;
    for (bool input : inputs)
    {
        result |= input;
    }
    return !result;
}
```

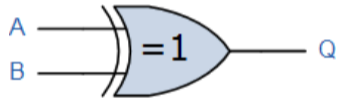
Adobe Stock
#319172794

Output

```
**********************************************
Select the gate:
1. AND
2. OR
3. NOT
4. XOR
5. XNOR
6. NAND
7. NOR
**********************************************
Enter the number corresponding to the gate: 7
**********************************************
Enter the number of inputs: 3
Truth Table:
0 0 0 | 1
1 0 0 | 0
0 1 0 | 0
1 1 0 | 0
0 0 1 | 0
1 0 1 | 0
0 1 1 | 0
1 1 1 | 0
**********************************************
```

XOR GATE

XOR gate: The output is 1 (true) when an odd number of its inputs are set to 1. In other words, if an even number of inputs are 1, the output is 0 (false)

Symbol	Truth Table		
 2-input Ex-OR Gate	A	B	Q
	0	0	0
	0	1	1
	1	0	1
	1	1	0

Function to perform logical XOR

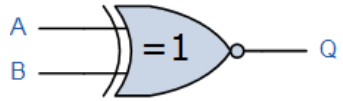
```cpp
// Function to perform logical XOR operation
bool logicalXOR(const vector<bool>& inputs)
{
    bool result = false;
    for (bool input : inputs)
    {
        result ^= input;
    }
    return result;
}
```

Output

```
**********************************************
Select the gate:
1. AND
2. OR
3. NOT
4. XOR
5. XNOR
6. NAND
7. NOR
**********************************************
Enter the number corresponding to the gate: 4
**********************************************
Enter the number of inputs: 3
Truth Table:
0 0 0 | 0
1 0 0 | 1
0 1 0 | 1
1 1 0 | 0
0 0 1 | 1
1 0 1 | 0
0 1 1 | 0
1 1 1 | 1
**********************************************
```

XNOR GATE

XNOR gate: The output is 1 (true) when an even number of its inputs are set to 1. In other words, if an odd number of inputs are 1, the output is 0 (false)

Symbol	Truth Table		
 2-input Ex-NOR Gate	A	B	Q
	0	0	1
	0	1	0
	1	0	0
	1	1	1

Function to perform logical XNOR

```cpp
// Function to perform logical XNOR operation
bool logicalXNOR(const vector<bool>& inputs)
{
    int trueCount = 0;
    for (bool input : inputs)
    {
        if (input)
            trueCount++;
    }
    return (trueCount % 2 == 0);
}
```

Output

```
1. AND
2. OR
3. NOT
4. XOR
5. XNOR
6. NAND
7. NOR
**********************************************
Enter the number corresponding to the gate: 5
**********************************************
Enter the number of inputs: 4
Truth Table:
0 0 0 0 | 1
1 0 0 0 | 0
0 1 0 0 | 0
1 1 0 0 | 1
0 0 1 0 | 0
1 0 1 0 | 1
0 1 1 0 | 1
1 1 1 0 | 0
0 0 0 1 | 0
1 0 0 1 | 1
0 1 0 1 | 1
1 1 0 1 | 0
0 0 1 1 | 1
1 0 1 1 | 0
0 1 1 1 | 0
1 1 1 1 | 1
**********************************************
```

Main Function

```
int main()
{
    bool continueGenerate = true;
    while (continueGenerate)
    {
        cout << "************************************************" << endl;
        cout << "Select the gate: " << endl;
        cout << "1. AND" << endl;
        cout << "2. OR" << endl;
        cout << "3. NOT" << endl;
        cout << "4. XOR" << endl;
        cout << "5. XNOR" << endl;
        cout << "6. NAND" << endl;
        cout << "7. NOR" << endl;
        cout << "************************************************" << endl;
        cout << "Enter the number corresponding to the gate: ";
        int gateChoice;
        cin >> gateChoice;
        cout << "************************************************" << endl;

        while (gateChoice < 1 || gateChoice > 7)
        {
            cout << "Invalid choice. Please enter a valid choice." << endl;
            cout<<"Enter the number corresponding to the gate: ";
            cin >> gateChoice;
            cout << "************************************************" << endl;
        }
```

Main Function (cont)

```
cout << "Enter the number of inputs: ";
int numInputs;
cin >> numInputs;

while (numInputs < 1)
{
    cout << "Invalid number of inputs. Please enter a valid number." << endl;
    cout << "Enter the number of inputs: ";
    cin >> numInputs;
}

int numOutputs = 1;
vector<bool> inputs(numInputs);
int numCombinations = pow(2, numInputs);

cout << "Truth Table:" << endl;
for (int i = 0; i < numCombinations; i++)
{
    // Generate all possible input combinations
    for (int j = numInputs - 1; j >= 0; j--)
    {
        inputs[j] = i & (1 << j);
    }

    bool output = false;
```

Main Function (cont)

```
bool output = false;
switch (gateChoice)
{
case 1:
    output = logicalAND(inputs);
    break;
case 2:
    output = logicalOR(inputs);
    break;
case 3:
    output = logicalNOT(inputs[0]);
    break;
case 4:
    output = logicalXOR(inputs);
    break;
case 5:
    output = logicalXNOR(inputs);
    break;
case 6:
    output = logicalNAND(inputs);
    break;
case 7:
    output = logicalNOR(inputs);
    break;
}
```

Main Function (cont)

```
            // Print input and output values for this combination
            for (bool input : inputs)
            {
                cout << input << " ";
            }
            cout << "| " << output << endl;
        }

        char continueGenerateChoice;
        cout << "*********************************************" << endl;
        cout << "Do you want to continue? (Y/N): ";
        cin >> continueGenerateChoice;
        if (toupper(continueGenerateChoice) != 'Y')
        {
            break; // Exit the loop if the user does not want to continue
        }
    }
    cout << "Exiting the program.......";
    return 0;
}
```


GOOD BYE